$T(v) = Av$

$A \in M_{n \times n}$

Def. $v \in \mathbb{R}^n$ is an <u>eigenvector</u> with <u>eigenvalue</u> $\lambda \in \mathbb{R}$

if (i) $v \neq 0$

and (ii) $Av = \lambda v$

Def. The <u>λ-eigenspace</u> of A is

$\{ v \in \mathbb{R}^n : Av = \lambda v \}$ = eigenvectors of λ and $0 \in \mathbb{R}^n$

Example

Take $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$ and take for granted

that 2 is an eigenvalue of A. Then

$$\begin{aligned} 2\text{-eigenspace} &= \mathrm{Nul}(A - 2I) \\ &= \mathrm{Nul}\left(\begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix} \right) \\ &= \text{solution to } \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

REF of $\begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$ is $\begin{bmatrix} 1 & -1/2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

So $\mathrm{Nul}(A - 2I) = \left\{ c_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} : c_2, c_3 \in \mathbb{R} \right\}$

$= \mathrm{span}\left\{ \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

Theorem $\Rightarrow \left\{ \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for the 2-eigenspace.

Since there are two vectors in this basis the 2-eigenspace of A is 2-dimensional.

Examples

Suppose v is an eigenvector of A with eigenvalue λ.

$$\Rightarrow \quad Av = \lambda v.$$

$$\Rightarrow \quad A\,Av = A(\lambda v)$$

$$\Rightarrow \quad A^2 v = \lambda A v$$

$$\Rightarrow \quad A^2 v = \lambda \lambda v$$

$$\Rightarrow \quad A^2 v = \lambda^2 v \qquad (*)$$

In conclusion $v \neq 0$ is an eigenvector for A^2 with eigenvalue λ^2.

Multiplying $(*)$ by A on the left gives

$A^3 v = \lambda^3 v$ and v is an eigenvector of A^3 with eigenvalue λ^3.

By induction on $k \geq 1$, v is an eigenvector of A^k with eigenvalue λ^k.

Recognizing eigenvectors.

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Similarly given a diagonal matrix

$$\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \lambda_1\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} e_1 = e_1$$

Generally $\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} e_j = \lambda_j e_j$ for the standard basis $\{e_1, \dots, e_n\}$.

Example

$$\begin{bmatrix} 2 & -4 \\ -2 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2-4 \\ -2-0 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix} = -2\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Generally if the sum of the values of every row of a matrix is λ then $\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$ is an eigenvector with eigenvalue λ.

Theorem

$A \in M_{n \times n}$ is invertible $\iff$ 0 is <u>not</u> an eigenvalue of A.

Sketch of proof.

A is invertible $\iff$ $Ax = 0$ only has $x = 0$ as a solution (Thm 8 §2.3)

$\Leftrightarrow$ $Ax = 0x$ only has $x=0$ as a solution

$\Leftrightarrow$ 0 is not an eigenvalue of A.

Theorem 2

Suppose $v_1, \dots, v_k$ are eigenvectors of A with eigenvalues $\lambda_1, \dots, \lambda_k$ respectively ($A v_j = \lambda_j v_j$ $\forall$ $1 \le j \le k$)

If $\lambda_1, \dots, \lambda_k$ are distinct ($\lambda_j \neq \lambda_\ell$ for $j \neq \ell$)

then $\{v_1, \dots, v_k\}$ is lin. ind.

A consequence of Thm 2 is that the lin. ind. set $\{v_1, \dots, v_k\}$ is a basis of $\text{span}\{v_1, \dots, v_k\} \subseteq \mathbb{R}^n$.

$\text{span}\{v_1, \dots, v_k\}$ is the k-dimensional.

Example

Take $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

Clearly $Av = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} v = 3Iv = 3v$ $\forall v \in \mathbb{R}^2$.

$\Rightarrow$ Every $v \neq 0$ is an eigenvector with eigenvalue 3.

Theorem 2 does not apply since we only have a single eigenvalue, namely 3.

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§5.2 The Characteristic Equation

In this section $A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \in M_{n\times n}$

Def. $\det(A) = (-1)^{1+1} a_{11} \det A_{11} + (-1)^{2+1} a_{12} \det A_{12} + \cdots$

$$+ (-1)^{n+1} a_{1n} \det A_{1n}$$

where A_{1j} is the matrix obtained from A by removing the first row and the j'th column.

If $A \in M_{1\times 1}$ then $\det(A) = \det[a_{11}] = a_{11}$.

Theorem $\det(A) = (-1)^{j+1} a_{j1} \det A_{j1} + \cdots + (-1)^{j+n} a_{jn} \det(A_{jn})$

expansion along j'th row

$= (-1)^{j+1} a_{1j} \det A_{1j} + \cdots + (-1)^{j+n} a_{nj} \det A_{nj}$

expansion along j'th column

Example

$$\det \begin{bmatrix} 2 & 3 & 1 \\ 4 & 0 & -1 \\ 0 & 2 & 1 \end{bmatrix} = (-1)^{1+1} 2 \det \begin{pmatrix} 0 & -1 \\ 2 & 1 \end{pmatrix} + (-1)^{2+1} 3 \det \begin{pmatrix} 4 & -1 \\ 0 & 1 \end{pmatrix}$$

$$+ (-1)^{3+1} 1 \det \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= 2(2) - 3(4) + 8$$

$$= 4 - 12 + 8$$

$$= 0$$

$$\det\left(\begin{pmatrix} 2 & 3 & 1 \\ 4 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix}\right) = (-1)^{3+2} 2 \det \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix} + (-1)^{3+3} 1 \det \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}$$

$$= \cdots = 0$$

Theorem 3

Suppose $A, B \in M_{n\times n}$. Then

(a) A is invertible $\iff \det(A) \neq 0$.

(b) $\det(AB) = \det(A)\det(B)$

(c) $\det(A^T) = \det(A)$ where A^T is the transpose of A.

(the jk entry of A^T is a_{kj})

(d) If A is upper (or lower) triangular then $\det(A)$ is the product of its diagonal entries

$$\det(A) = a_{11}a_{22}\cdots a_{nn}$$

Theorem

$\lambda \in \mathbb{R}$ is an eigenvalue of $A \iff \det(A-\lambda I)=0$.

Sketch of Proof.

$$\begin{aligned} Av = \lambda v &\iff Av = \lambda I v \\ &\iff Av - \lambda I v = 0 \\ &\iff (A-\lambda I)v = 0 \qquad (v \neq 0) \\ &\iff A - \lambda I \text{ is not invertible} \quad (\text{Thm 8 §2.3}) \\ &\iff \det(A-\lambda I) = 0 \quad (\text{Thm 3}) \end{aligned}$$

The theorem tells us that λ is an eigenvalue $\iff$ λ is a solution to $\det(A - xI) = 0$

Def. The polynomial $\det(A-xI)$ is called the characteristic polynomial of A. The equation

$\det(A - xI) = 0$ is called the <u>characteristic equation</u> of A.

Example

Take $A = \begin{bmatrix} 5 & -2 & 6 \\ 0 & 3 & -8 \\ 0 & 0 & 5 \end{bmatrix}$

Then $A - xI = \begin{bmatrix} 5 & -2 & 6 \\ 0 & 3 & -8 \\ 0 & 0 & 5 \end{bmatrix} - \begin{bmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & x \end{bmatrix}$

$$= \begin{bmatrix} 5-x & -2 & 6 \\ 0 & 3-x & -8 \\ 0 & 0 & 5-x \end{bmatrix}$$

$$\det(A - xI) = (5-x)(3-x)(5-x) \quad \text{(since upper triangular)}$$

$$= (-(x-5))(-(x-3))(-(x-5))$$

$$= -(x-5)^2(x-3)$$

$$\det(A - xI) = 0 \iff -(x-5)^2(x-3) = 0$$

$$\iff (x-5) = 0 \text{ or } x-3 = 0$$

$$\iff x = 5 \text{ or } x = 3.$$

Therefore A has two eigenvalues $\lambda = 3, 5$.

Theorem 1 (§ 5.1)

If A is triangular then the eigenvalues of A are the diagonal entries $a_{11}, a_{22}, \ldots, a_{nn}$